

Charugundla Chethan

Third Year Undergraduate, Computer Science and Engineering, Indian Institute of Technology Kanpur
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ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech, CSE	Indian Institute of Technology Kanpur	8.14/10
2021	Andhra Pradesh Board (XII)	Bhashyam Junior College, Guntur	97.1%
2019	Andhra Pradesh Board (X)	Bhashyam Public School, Dharmavaram	10/10

SCHOLASTIC ACHIEVEMENTS

- Achieved a perfect **100 Percentile** in Mathematics subject during the **JEE Mains 2021** organized by the **NTA**
- Secured **All India Rank 891** in common rank list in **JEE Advanced 2021** among **1.4 Lakh** shortlisted candidates
- Secured **All India Rank 489** in common rank list in **JEE Mains 2021** among **9.4 Lakh** appeared candidates
- Secured **Rank 58** among **1.5 Lakh** appeared candidates in **TS Eamcet 2021** conducted by state board of Telangana
- Secured **Rank 99** among **1.7 Lakh** appeared candidates in **AP Eamcet 2021** conducted by state board of Andhra Pradesh

COMPETITIVE PROGRAMMING ACHIEVEMENTS

- Achieved highest rating of **2061 (5 Stars)** on **CodeChef** and **Global Rank 6** in **CodeChef Starters 120 Division 2**
- Achieved highest rating of **1858 (Expert)** on **CodeForces** and **Global Rank 329** in **CodeForces Round 917 Div 2**
- Achieved highest rating of **2065 (Knight)** on **LeetCode** and highest rating of **1200 (4 Kyu)** on **AtCoder**

KEY PROJECTS

- **Mess Automation System** (Jan'23-Apr'23)
Course Project: CS253, Professor Indranil Saha, Department of Computer Science and Engineering, IIT Kanpur
 - Developed a **Web Application** for IITK students, that aims to automate various operations of a hostel mess
 - Involved in **team management of 8**, and in **design** and **testing** and gave active feedback to the **development team**
 - Utilized **HTML, CSS, Java script, Bootstrap** and **EJS** Templates for Frontend and **Node.Js** for Backend
 - Utilized **MongoDB** for efficient Database management and seamlessly collaborated through **GitHub** 📄
 - Contributed in documentation of Software Requirements Specification Document, Software Design Document, Software Implementation Document, Testing Document and User Manual of our software
- **CSE Bubble Processor** (Mar'23-Apr'23)
Course Project: CS220, Professor Urbi Chaterjee, Department of Computer Science and Engineering, IIT Kanpur
 - Built **CSE Bubble processor** having **Instruction Set Architecture (ISA)** similar to **MIPS architecture** implementing a streamlined **Single-Cycle Instruction Fetch, Decode, and Execution** approach
 - Designed and implemented the **Arithmetic Logic Unit (ALU)** in **top-down approach** to develop different modules for **R-type, I-type, J-type** instructions and the **Finite State Machine** for the **control signals** to execute the CSE Bubble
 - Developed **MIPS Code** for **Bubble Sort** algorithm, translated to **machine code** following **ISA** and executed the processor
- **Cage of Algo** (Jun'22-Jul'22)
Club Project: Programming Club, Science and Technology Council, IIT Kanpur
 - Learnt about Fundamental Concepts of Competitive Programming including Time and Space Complexity, STL in C++
 - Learnt Intermediate Concepts including Binary Search, Sorting Algorithms, Dynamic Programming and Range Queries
 - Acquired knowledge about Trees including Binary Search Tree and Graph Algorithms including Breadth First Search, Depth First Search, Disjoint Set Union, Topological sort, Kruskal Algorithm and Dijkstra Algorithm
 - Actively participated in practice contests organized by project mentors, designed to focus on specific topic-based questions

TECHNICAL SKILLS

Web Development	Programming Languages	Utilities and Tools
HTML, CSS, EJS, Node.Js, MongoDB	C, C++, Verilog HDL, MIPS	Git, Bash, L ^A T _E X, AutoCAD

RELEVANT COURSES

Data Structure and Algorithms Computer Organisation Introduction to Probability	Fundamentals of Computing Introduction to Logic Introduction to Electronics	Software Development and Operations Discrete Mathematics Linear algebra and Differential Equations
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OTHER INTERESTS

- Highly interested in **competitive programming**, actively expanding knowledge and skills in algorithms, data structures, and problem-solving techniques